CLINICAL · EAR

Labyrinthine fistula and labyrinthitis

A dizzy patient with an infected ear raises one question: has the labyrinth been breached, or inflamed? A fistula can be sealed, and serous labyrinthitis recovers; suppurative labyrinthitis destroys the hearing for good. After meningitis, the cochlea can turn to bone within weeks, and the window for a cochlear implant can close.

AT A GLANCE

- Vertigo in an infected ear means the labyrinth is breached or inflamed.
- Serous disease can recover; suppurative disease causes a dead ear.
- Fistula test is specific but insensitive.
- Treat the source; add steroids and follow hearing after meningitis.
- Early implant if ossification threatens.

1 CHAPTER ONE Overview

When infection reaches the inner ear.

1.1 Definitions

1.1.1 Labyrinthine fistula

Erosion of the bony labyrinth, usually the lateral semicircular canal, exposing the membranous labyrinth without infecting it. Dhingra calls it *circumscribed labyrinthitis*.

1.1.2 Serous labyrinthitis

Diffuse irritation of the labyrinth by bacterial toxins through the windows or a fistula, without bacteria in the inner ear. Reversible.

1.1.3 Suppurative labyrinthitis

Bacteria and pus in the labyrinth: total, permanent loss of hearing and vestibular function.

1.1.4 Perilymph fistula

> A leak of perilymph through the oval or round window, after stapes surgery, barotrauma, diving, straining or trauma.

1.2 Other forms

Viral labyrinthitis (mumps, measles, rubella, cytomegalovirus) and syphilitic labyrinthitis.

2 CHAPTER TWO Pathways

How infection gets in.

2.1 From the middle ear

Cholesteatoma eroding the lateral canal (the commonest), through the oval or round window in acute or chronic otitis media, and after surgery or trauma.

2.2 From the meninges

Bacteria spread from the subarachnoid space through the cochlear aqueduct or the internal auditory canal. *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Neisseria meningitidis* are the organisms; pneumococcus carries the highest risk of deafness, around 20%.

2.3 Blood-borne

Viral and syphilitic infections.

3 CHAPTER THREE Labyrinthine fistula

The eroded lateral canal.

3.1 Causes

Cholesteatoma in most cases, then tumours, trauma and surgery. About 7 to 10% of cholesteatomas have a fistula, 90% of them in the lateral canal.

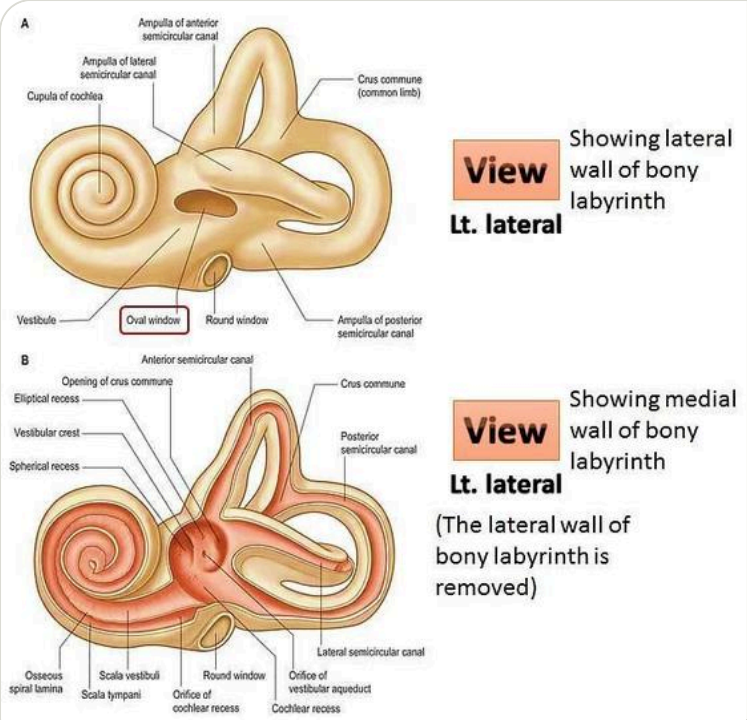


FIGURE 1 The bony labyrinth: the lateral semicircular canal bulges into the medial wall of the attic and antrum, where cholesteatoma erodes it.

Anatomy atlas · ENT Steps p. 64

3.2 Clinical picture

Brief vertigo provoked by pressure on the ear: pressing the tragus, cleaning the ear, or nose blowing. Hearing is often preserved.

3.3 Fistula test

Pressure applied to the canal (tragal pressure or a Siegle speculum) moves the endolymph through the fistula and causes vertigo and nystagmus. With positive pressure the fast phase usually beats towards the tested ear; with negative pressure, away from it.

THE FISTULA TEST

It is specific but insensitive: positive in only 30 to 50% of fistulae. A negative test does not exclude one; it may also be negative in a dead labyrinth or if cholesteatoma seals

the fistula. Assume a fistula in every cholesteatoma.

3.4 Other positive fistula tests

Ménière's disease (Hennebert's sign), superior canal dehiscence, perilymph fistula and congenital syphilis.

3.5 Investigations

HRCT shows only about 60% of fistulae before surgery.

3.6 Management

Surgery for the cholesteatoma; the matrix over a small fistula may be removed carefully at the end of the operation and the defect sealed with fascia or bone pâté; over a large fistula, or in an only hearing ear, the matrix is left and the ear exteriorised.

4 CHAPTER FOUR Serous and suppurative labyrinthitis

Irritation or destruction.

4.1 Serous labyrinthitis

4.1.1 Clinical picture

Vertigo with nausea, nystagmus towards the affected ear (an irritative lesion), and some sensorineural hearing loss that recovers if treated.

4.1.2 Treatment

Bed rest with the affected ear uppermost, full-dose antibiotics, labyrinthine sedatives, and myringotomy for a bulging drum. Mastoidectomy treats a chronic source once medical treatment has begun.

4.2 Suppurative labyrinthitis

4.2.1 Clinical picture

Severe vertigo with nausea and vomiting, nystagmus beating towards the healthy ear (the labyrinth has failed), total deafness in the ear, and a toxic patient. The vertigo settles over three to six weeks as the brain compensates.

4.2.2 Danger

Infection can spread from the labyrinth to the meninges along the internal auditory canal.

4.2.3 Treatment

Intravenous antibiotics, removal of the middle ear source, and rarely drainage of the labyrinth if it is feeding an intracranial infection.

SEROUS OR SUPPURATIVE?

	SEROUS	SUPPURATIVE
Cause	Toxins	Bacteria
Nystagmus	Towards the affected ear	Towards the healthy ear
Hearing	Partial loss, recovers	Total, permanent loss
Patient	Unwell	Toxic, may have meningitis
Outcome	Recovery	Dead ear, ossification

4.3 Labyrinthitis ossificans

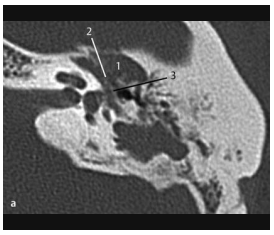
After suppurative labyrinthitis, especially meningitic, fibrous tissue and new bone fill the cochlea, starting in the basal turn near the round window, within weeks to months.

DEAFNESS AFTER MENINGITIS

- Test hearing in every patient after bacterial meningitis.
- Bilateral profound loss: image urgently (MRI and CT) for ossification and refer for early cochlear implantation.
- Steroids reduce hearing loss in *H. influenzae* and pneumococcal meningitis.

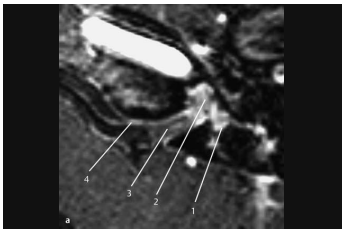
IMAGING IN THE APP 4 cases

Labyrinthine fistula, labyrinthitis ossificans, labyrinthitis with meningitis, and otosyphilis.



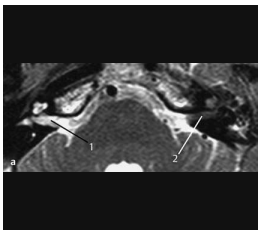
Labyrinthine Fistula due to Cholesteatoma

Infection & inflammation · 11 images



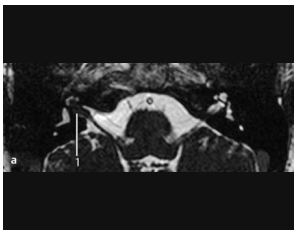
Labyrinthitis Ossificans (Ossification after Meningitis)

Bone disease · 14 images



Labyrinthitis and Subclinical Meningitis

Infection & inflammation · 5 images



Otosyphilis and Intracranial Complications

Infection & inflammation · 6 images

OPERATIVE GUIDES IN THEATRE 3 operations

Mastoid Surgery
Remove source

Cochlear Implantation in Adults
Implant before ossification

Pediatric Cochlear Implantation
Paediatric implant

5 CHAPTER FIVE Perilymph fistula

A leak at the windows.

5.1 Causes

Stapes surgery, barotrauma, diving, heavy lifting, straining or coughing, and head injury. A leak in an unoperated ear usually sits around the annular ligament or the round window membrane. Spontaneous fistula is controversial.

5.2 Clinical picture

Fluctuating or sudden hearing loss, roaring tinnitus, disequilibrium or vertigo, often after the precipitating event, and a positive fistula test (Hennebert's sign). A penetrating ear injury with vertigo and hearing loss suggests stapes subluxation into the vestibule.

5.3 Management

Bed rest for 7 to 10 days with the head raised, no straining, and stool softeners. Explore and patch the window if hearing is worsening or symptoms persist.

6 CHAPTER SIX Viral and syphilitic labyrinthitis

The other causes.

6.1 Viral

Mumps (usually one-sided profound loss), measles, rubella and cytomegalovirus. CMV is the commonest non-syndromic cause of congenital sensorineural hearing loss, and the loss may progress after a normal newborn screen.

6.2 Syphilis

Congenital or acquired. Late congenital syphilis may first appear at 8 to 20 years as bilateral sudden or fluctuating loss with vertigo, and Hennebert's sign. Test with VDRL or RPR and FTA-ABS; treat with penicillin and steroids.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 On call

VERTIGO IN AN INFECTED EAR

1. Examine for nystagmus and its direction; do a gentle fistula test.
2. Audiogram and HRCT.
3. Intravenous antibiotics; myringotomy for acute otitis media.
4. Plan surgery for cholesteatoma.
5. Hearing loss with meningitis: treat urgently and image for ossification.

7.2 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Three types:** circumscribed, serous, suppurative.
- **The fistula test,** its direction, and why it can be negative.
- **Nystagmus direction** in serous versus suppurative labyrinthitis.
- **Labyrinthitis ossificans** and early cochlear implantation.
- **Perilymph fistula:** causes and treatment.

7.3 Common pitfalls

COMMON PITFALLS

- Being reassured by a negative fistula test.
- Forgetting a hearing test after meningitis.

- Missing late congenital syphilis in a young adult with fluctuating bilateral loss.

Written for this app from Oxford Textbook of Otolaryngology ch. 7, Oxford Specialist Handbook ch. 18, Dhingra ch. 7, 12 and K. J. Lee ch. 22, 23. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

MIDDLE EAR & MASTOID · INFECTION & INFLAMMATION

Labyrinthine Fistula due to Cholesteatoma

IMAGING FINDINGS

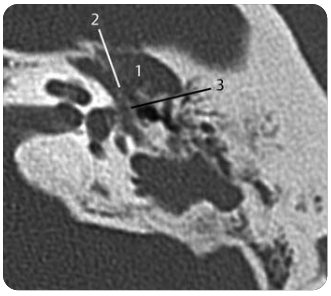
- Loss of the bony cover of the lateral semicircular canal next to cholesteatoma
- Best seen on coronal and Pöschl or Stenvers reformats
- Labyrinthine fluid on T2 may be reduced if labyrinthitis follows

TOP DIFFERENTIALS

- Partial volume of thin normal bone
- Superior canal dehiscence
- Post-surgical defect

CHECKLIST

- Report fistula site and size before surgery
- Confirm on thin slices in two planes



CT, axial. Note the smooth-bordered opacification suggestive of cholesteatoma in the anterior part of the middle ear (1), at the level of the…



Coronal bone CT reveals a small pars flaccida cholesteatoma filling the Prussak space with blunting of the scutum. The head of the malleus is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050.

INNER EAR & IAC · BONE DISEASE

Labyrinthitis Ossificans (Ossification after Meningitis)

IMAGING FINDINGS

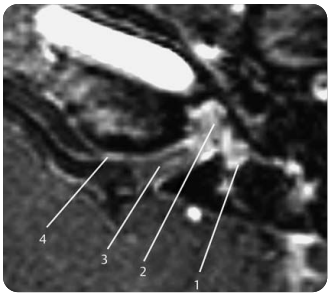
- Loss of the bright fluid signal of the cochlea on high-resolution T2
- Later: bone in the basal turn and canals on CT
- Early: enhancement of the labyrinth

TOP DIFFERENTIALS

- Labyrinthine or cochlear aplasia
- Cochlear otosclerosis
- Intralabyrinthine schwannoma

CHECKLIST

- After meningitis: urgent MRI and cochlear implantation
- Report patency of the basal turn



MRI, T1-weighted with gadolinium enhancement, axial. There is enhancement of the vestibule (1), cochlear contents (2), and the…



Axial bone CT in a 2-year-old boy with bilateral sensorineural hearing loss

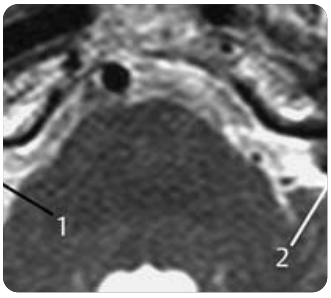
Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1116.

INNER EAR & IAC · INFECTION & INFLAMMATION

Labyrinthitis and Subclinical Meningitis

- Otoscopy may exclude middle ear infections although CT may be indicated in some cases. MRI will show (retro)cochlear enhancement in cases of (acute) infections.
- Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and ossification. In bilateral cases, this may hamper insertion of a cochlear implant.

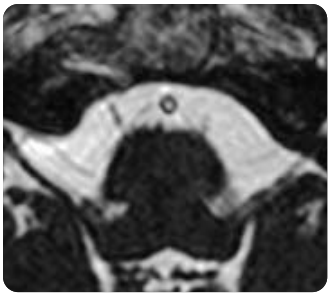


MRI, T2-weighted, axial. Note the normal fluid content of the internal auditory canal on the right side (1), whereas on the affected side, the…

INNER EAR & IAC · INFECTION & INFLAMMATION

Otosyphilis and Intracranial Complications

- Cranial nerve deficits, rapid progression, and multiple lesions are strongly indicative of inflammatory disorders. If malignancy is suspected, beware of involvement of other organs as primary sites of possible metastases.
- Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and/or ossification. In bilateral cases, this may hamper insertion of a cochlear implant, although its success may also be restricted by retrocochlear deficits.



Two weeks prior to this MRI investigation, this young man presented with sudden deafness of the right ear, followed by severe…